

SIH26091

Final Completion Report

West Bengal Hyper-Local Economic Network Repair

Version 0.3.0 | Audited 28 August 2026

Evidence-first implementation, data-freshness audit, real survey model training, production integration, multi-district validation, mathematical audit and honest decision limitations.

Outcome

The product is a working West Bengal evidence, spatial-network, optimization and finance-screening system with real survey processing, trained model artifacts, geographic holdout metrics, production survey priors, HTTP UI, and fail-closed decision gates. It does not issue a 2026 venture recommendation when current business-critical evidence is unavailable.

Exact geographic coverage

- Operational identity store: 53,537 West Bengal village/town/ward identities.
- Census 2011 Location Code Directory: 41,154 West Bengal rows from state code 19.
- Census PCA: all 19 West Bengal districts as defined in 2011; 179,148 population/household/sex

evidence records.

- Livestock: 40,475 source identities and 202,375 species records across all 23 publisher district

labels present in the West Bengal extract.

- OSM: statewide regional extract with 633,601 road ways, 17,212 POIs/places and 381 admin areas.
- HCES/ASUSE: district/sector survey priors cover 23 current-survey district groups. They are

regional sampled estimates and are not exact locality observations.

Statewide geography and routing coverage must not be read as statewide economic completeness.

Datasets, periods and freshness

Dataset / variable	Period	Status	Permitted decision use
Census geography and PCA	2011	HISTORICAL_BASELINE	Structural identity and historical population only.
20th Livestock Census	2019	STALE_FOR_DECISION	Historical stock context; not current reachable milk supply.
HCES	2022-23	STALE_FOR_DECISION	Comparison wave only.
HCES	Aug 2023-Jul 2024	RECENT	District/rural-urban consumption prior; never exact village demand.
ASUSE	Oct 2023-Sep 2024	historical comparison	Aggregate comparison priors.
ASUSE	Jan-Dec 2025	RECENT	District/sector/NIC2 enterprise prior; not competitor capacity.
BAHS	2025 publication	RECENT state context	State yield context only; not locality supply.
OSM publisher snapshot	retrieved 2026-08-27	CURRENT structural proxy	Catchments/routing/POI proxies with completeness caveat.
PMMY	Tarun Plus effective 2024-10-24; page updated 2026-02-05	CURRENT	Category/sector/collateral screening only.
AHIDF / IDF	temporary continuation 2026-04-01 to 2026-09-30	CURRENT	Conditional scheme screen; portal/lender confirmation required.
Current local selling/procurement price	unavailable	UNKNOWN	Blocking.
Fuel/transport cost and incumbent capacity	unavailable	UNKNOWN	Blocking.
CAPEX/OPEX, wage and rent	unavailable	UNKNOWN	Blocking.
Current locality population/projection	unavailable	UNKNOWN	Blocking; no 2011 value is relabelled as 2026.
Current lender rate/tenure/underwriting	lender-specific and unavailable	UNKNOWN	Blocking.

No 2026 population projection is stored. Therefore no projection method, central/lower/upper value or uncertainty is claimed.

Restricted raw archives

Archive	Bytes	SHA-256
HCES 2022-23 CSV	244,317,752	093a51d337eb07eede7d6e2a00ec55790d926bacf869be2cf2825abd629444ae
HCES 2023-24 CSV	256,315,433	acf7b9cc840676fb812c05c48f09fa034955fe2cad5112e6d9b6d852f5f2e267
ASUSE 2023-24 CSV	98,219,217	82a2e59aab71fee6dda7cd5f859c7cfbcf5e23ddf70b36a5a2962296b838de54
ASUSE calendar-2025 CSV	173,738,167	f20b35abdb7ba97daaca24f4b927a277e95e1145fcb86b2bd97304f771dc591

All four passed full member CRC checks. They remain applicant-only and are not included in the public-share package.

Models actually trained and validated

Validation is leave-one-district-out across 23 West Bengal district groups.

HCES household liquid-milk target

- Rows: 18,120; zero targets: 7,560.
 - Target: zero-inclusive monthly liquid-milk litres per household member.
 - Weighted category-mean baseline: MAE 1.384585; RMSE 1.748752.
 - Ridge one-hot: MAE 1.367413; RMSE 1.733946.
 - Random forest: MAE 1.353607; RMSE 1.724402; weighted bias 0.000133;
- mean calibration ratio 1.000082; decile calibration MAE 0.131882.

- Holdout winner: random forest.

ASUSE enterprise annual-GVA target

- Rows: 38,626; zero targets: 40.
 - Target: annualized enterprise GVA in INR using official item 769 and reference-period rule.
 - Weighted category-mean baseline: MAE INR 97,124.15; RMSE INR 259,004.56.
 - Ridge one-hot: MAE INR 68,004.76; RMSE INR 197,753.34.
 - Random forest: MAE INR 61,573.20; RMSE INR 193,228.67; weighted bias INR 22.54;
- mean calibration ratio 1.000151; decile calibration MAE INR 2,528.68.

- Holdout winner: random forest.

The fitted random forests won the comparison, but both ML inference paths are withheld from normal locality requests because those requests lack the household/enterprise microfeatures used in training. Production uses the direct weighted district/sector or district/sector/NIC2 survey estimator. This is a deliberate applicability decision, not a claim that no model was trained.

Production algorithm and MVV status

Ordinary flow is locality → stored evidence and regional priors → freshness gates → demand/supply/ price/capacity → OSM catchment → graph → exact min-cost maximum flow → bottleneck → automatic candidate → exact finite-candidate MVV → finance screen → digital twin/stress when assumptions exist.

- Flow: exact maximum served demand, then minimum economic cost.
- Counterfactual: newly served demand and cannibalized incumbent flow separated.
- MVV: exact exhaustive oracle over supplied candidates; minimum investment with demand/income/

capital constraints.

- Automatic library: one dairy rented-transport primitive when all source-linked costs/capacities

exist. No general configuration MILP or complete venture library is claimed.

- Uncertainty: survey confidence intervals and endpoint multiplication are retained. Full correlated probabilistic propagation is not implemented.

- Robustness: demand failure boundary and exact minimax regret over an explicit finite table.

Finance rules and effective dates

- PMMY: official Department of Financial Services page updated 2026-02-05; Tarun Plus up to INR

20 lakh for successful prior Tarun borrowers, effective 2024-10-24. No universal lender rate, tenure or sanction is inferred.

- AHIDF/IDF: temporarily continued from 2026-04-01 through 2026-09-30 or earlier superseding

approval. Existing terms include 3% interest subvention, lender finance up to 90% of eligible cost, and up to eight years including two-year principal moratorium. These are ceilings, not promised borrower terms. Live portal, margin, gross/net rate, security and sanction remain gates.

Real E2E and browser validation

Tested localities: Kolkata; Amdanga (North 24 Parganas); Abad Bhagabanpur (South 24 Parganas); Kharibari CT (Darjeeling); Anandapur (Jalpaiguri); Adina (Maldah); Adra CT (Purulia).

Each resolved an exact stored geo ID, attached relevant HCES/ASUSE priors, computed a 5 km OSM catchment and returned INSUFFICIENT_EVIDENCE with no graph-generated venture. The UI was also tested through HTTP in a real Chromium browser using Kolkata: search, evidence loading, freshness, analysis, finance and source rendering all worked; browser console warnings/errors were zero.

Final verification

- Ruff: pass.
- Pytest: 36 passed.
- Dataset registry: 53/53 files match byte size and SHA-256.
- Restricted ZIP CRC: 4/4 pass.
- SQLite integrity: pass.
- Model artifact integrity/load: pass.
- ASUSE output-input=GVA: 1,551 groups, max residual $3.725290298461914e-09$ INR.
- E2E truth gate: 7/7 cases safe.

Remaining evidence gates

The product cannot honestly make a real 2026 locality venture recommendation until current local population/projection, demand calibration, productive/reachable supply, selling/procurement price, incumbent capacity, transport cost, venture configuration cost, and lender terms are supplied. It therefore returns uncertainty and explicit gates rather than fabricating those values.